# CAP2012 – DATA STRUCTURES

## CONTINUOUS ASSESSMENT 3 (CA3)

### LeetCode Problem Solving using C/C++ and GitHub Submission

| | | | |
|---|---|---|---|
| **Course Code** | CAP2012 | **Assessment** | CA3 |
| **Course Title** | Data Structures | **Total Marks** | 50 |
| **Section** | D1P2551 | **Group** | Group 2 Only |
| **Allocation Date:** | | **Submission Date:** | |

**SECTION D1P2551 – GROUP 2 ONLY**

## 1. Objective

This assessment is designed to strengthen problem-solving skills, understanding of data structures, algorithmic thinking, C/C++ implementation and testing, and the use of GitHub to maintain and present programming work.

## 2. Assessment Requirements

- Solve the 16 LeetCode problems provided, using C or C++ as the programming language.
- The assigned set consists of 9 Easy, 5 Medium, and 2 Hard problems (9×1 + 5×2 + 2×3 = 25 marks).
- The assigned problems are aligned with Units I–IV of CAP2012: Basic Concepts, Arrays, Linked Lists, and Stacks & Queues.
- Write an individual C/C++ solution for each problem. Code must be logically correct, readable, and appropriately named.
- Test each solution using the examples provided on LeetCode and additional relevant test cases, including boundary cases where applicable.
- Upload all solutions to one GitHub repository created specifically for this assessment.
- Ensure that the repository is accessible to the evaluator. Submit the correct repository URL through the channel specified by the faculty.
- Be prepared to explain the problem, data structure/approach used, C/C++ implementation, time and space complexity, test cases, and any minor modification asked during the viva.

## 3. LeetCode Questions

The complete list of LeetCode questions for Section D1P2551 – Group 1 is given below. Attempt all assigned questions in C/C++. Do not substitute any question without prior approval.

| No. | LeetCode # | Level | Problem Name | Topics Covered | Unit |
|---|---|---|---|---|---|
| 1 | 509 | Easy | Fibonacci Number | Recursion, loops | IV |
| 2 | 9 | Easy | Palindrome Number | Loops, control structures, arithmetic | I |
| 3 | 1929 | Easy | Concatenation of Array | Arrays, traversal, array modification | II |
| 4 | 35 | Easy | Search Insert Position | Arrays, traversal, deletion | II |
| 5 | 58 | Easy | Length of Last Word | Arrays, merging | II |
| 6 | 206 | Easy | Reverse Linked List | Singly linked list, pointers, traversal | III |
| 7 | 83 | Easy | Remove Duplicates from Sorted List | Linked list, traversal, deletion | III |
| 8 | 20 | Easy | Valid Parentheses | Stack, PUSH/POP operations | IV |
| 9 | 155 | Easy | Min Stack | Stack, queue, FIFO operations | IV |
| 10 | 125 | Medium | Valid Palindrome | Arrays, traversal, array manipulation | II |
| 11 | 92 | Medium | Reverse Linked List II | Linked list, pointer manipulation | III |
| 12 | 61 | Medium | Rotate List | Linked list, traversal, link manipulation | III |

| No. | LeetCode # | Level | Problem Name | Topics Covered | Unit |
|---|---|---|---|---|---|
| 13 | 232 | Medium | Implement Queue using Stacks | Stack, PUSH, POP, TOP operations | IV |
| 14 | 622 | Medium | Design Circular Queue | Circular queue, FRONT, REAR, enqueue/dequeue | IV |
| 15 | 32 | Hard | Longest Valid Parentheses | Stack, parentheses processing | IV |
| 16 | 23 | Hard | Merge k Sorted Lists | Linked lists, traversal, merging | III |

## 4. GitHub Repository and File-Naming Guidelines

Create one GitHub repository using the following naming convention:

CAP2012_CA3_LeetCode_<RegistrationNumber>

Example: CAP2012_CA3_LeetCode_12345678

Organize the C/C++ source files into folders according to difficulty level, as shown below:

```
CAP2012_CA3_LeetCode_<RegistrationNumber>/
├── Easy/
│   ├── 01_LC509_FibonacciNumber_Easy.cpp
│   └── ...
├── Medium/
│   ├── 10_LC189_RotateArray_Medium.cpp
│   └── ...
└── Hard/
    ├── 15_LC032_LongestValidParentheses_Hard.cpp
    └── 16_LC023_MergeKSortedLists_Hard.cpp
```

Use the following naming convention for every C/C++ source file:

SNo_LC<ProblemNumber>_<ShortProblemName>_<Difficulty>.<c/cpp>

Use S. No. as given in the table (01, 02 … 16), padded to two digits. Use the official LeetCode problem number, padded to at least three digits where appropriate (for example, 020, 023, 088, 509), followed by a concise problem name and difficulty level. Use the .c extension for C solutions and .cpp for C++ solutions.

## 5. Steps to Create a Repository and Upload Solutions

1. **Sign in to GitHub:** Open https://github.com and sign in to your account. Create an account if you do not already have one.
2. Create a new repository: Select the + icon or choose New repository. Enter the repository name using this format: CAP2012_CA3_LeetCode_<RegistrationNumber>.
3. **Choose repository visibility:** Select Public unless the faculty has instructed otherwise. The evaluator must be able to access your repository.
4. **Create the repository:** Select Create repository to finish setting it up.
5. Create folders and files: Use Add file → Create new file and enter a path such as Easy/01_LC509_FibonacciNumber_Easy.cpp. You may also upload prepared C/C++ files from your computer using Add file → Upload files.
6. Add the C/C++ solution: Paste or upload the complete solution for the relevant LeetCode problem. Use clear indentation and include a short comment with the problem number and title.
7. Commit the file: Enter a meaningful commit message, such as “Add 01 LC509 Fibonacci Number solution,” and select Commit changes.
8. Repeat for all assigned questions: Upload all 16 C/C++ solutions into the correct Easy, Medium, and Hard folders. Check that each file name includes the LeetCode number and difficulty level.
9. Verify the repository: Open the repository and check that all C/C++ files are visible, correctly named, and accessible. Copy the repository URL.

10. **Submit the link:** Submit the GitHub repository URL through the platform or channel specified by the faculty and keep the link available for the viva.

## 6. Evaluation Scheme (Total: 50 Marks)

| Component | Marks | Evaluation Criteria |
|---|---|---|
| Coding and GitHub Submission | 25 | Correctness and completeness of solutions; appropriate data structure/algorithm selection; C/C++ implementation; testing and handling of relevant cases; code readability; repository organization and file naming; successful upload and repository accessibility. |
| Viva / Code Evaluation | 25 | The student must explain the problem statement, data structure/approach used, implementation, time and space complexity, and test cases for the questions solved and uploaded to GitHub. Follow-up questions or a small code modification may be asked. |

**Important:** Students are expected to write and understand their own code. The viva will be based on the questions solved and uploaded to the student's repository. Missing, inaccessible, substituted, or incorrectly organized submissions may affect the Coding and GitHub component.

## 7. Student Submission Checklist

- ✓ All 16 assigned LeetCode problems attempted in C/C++ (9 Easy, 5 Medium, 2 Hard)
- ✓ Repository follows the prescribed naming convention
- ✓ C/C++ source files follow the prescribed file-naming convention.
- ✓ Repository link is accessible to the evaluator
- ✓ Repository URL has been submitted through the faculty-specified channel
- ✓ Prepared to explain and discuss the solutions during the viva